Wave speed

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Learning target: Use speed, frequency and wavelength in a consistent relationship.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 12.*
+40 J; internal energy increases.

02. Worked example: explain the model
Two cycles of 3 m versus three cycles of 2 m; speed remains 6 m/s.

03. Read and interpret the model
Higher frequency has half the wavelength.

04. Apply the physics
8 m/s; 1 m.

05. Explain your reasoning
Hz/m=1/(s·m), not m/s; correct product f λ gives m/s.

06. Review: connect an earlier idea *Review: Unit 5, Day 3.*
Electromagnetic radiation can travel in a vacuum; conduction/convection require matter.

07. Exit check
15 m/s.

08. STRETCH: test the model
Examples (1 Hz, 12 m),(2 Hz, 6 m),(4 Hz, 3 m).

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01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 12.*
Carnot limit $\eta \approx 0.4$; $W_{max} \approx 280$ J. Irreversibility and losses reduce actual useful work.

02. Worked example: explain the model
Two cycles of 3 m versus three cycles of 2 m; speed remains 6 m/s.

03. Read and interpret the model
Higher frequency has half the wavelength.

04. Apply the physics
$\lambda = 0.80$ m; f 680 Hz.

05. Explain your reasoning
Hz/m=1/(s·m), not m/s; correct product f λ gives m/s.

06. Review: connect an earlier idea *Review: Unit 5, Day 3.*
Net thermal radiation transfers energy to the cooler radiative environment. Absolute zero is not required.

07. Exit check
30 m/s;0.60 m.

08. STRETCH: test the model
3 and 2 m; driven oscillations keep the source frequency across a stationary boundary.